Universidad Carlos III de Madrid

Bachelor in Aerospace Engineering

Mechanics Applied to Aerospace Engineering

Particle on Oscillating Loop

Laboratory Session 2

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1 Introduction

In this laboratory report, we will investigate the dynamics of a heavy point particle P of mass m that slides without friction on a circular wire of centre C and radius R [1]. The wire is on a vertical plane Oxy that rotates about the vertical axis Oy with a known angular velocity $\mathbf{\Omega}(t) = A\omega \sin(\omega t)\,\mathbf{j}$, and the position of P on the wire is given by the angle φ between $\overrightarrow{CP}$ and Ox (Fig. 1a). It is the same system as Problem 34 of the course [2], but here we solve it numerically.

The main difficulty is that the frame where the motion is simple, the one attached to the wire, is not inertial and does not rotate at a constant rate, so the centrifugal, Euler and Coriolis forces all act on the particle. Our objectives are to derive the equation of motion and the reaction of the wire, to integrate it in Matlab with `ode45` and explain the forces and the energy we obtain, to find an analytical criterion that tells if the particle oscillates or rotates on a fixed loop and check it with simulations, and to see how far we can trust a linearized equation around the lowest point of the wire, where a resonance appears.

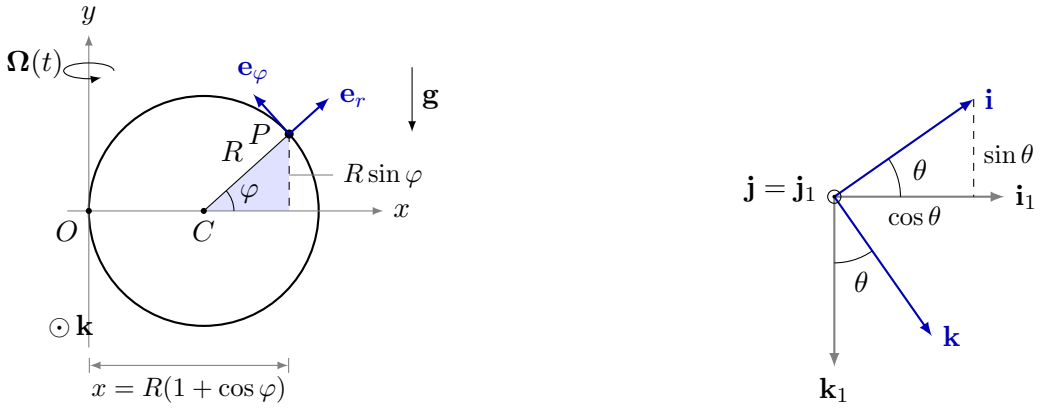
In Section 2 we derive the analytical model and in Section 3 we explain how we have implemented it in Matlab. In Section 4 we present the results and discuss each of them next to its figure, and Section 5 gathers the conclusions.

2 Analytical Model

2.1 Reference Frames and Degrees of Freedom

We use the notation of the guidelines [1]. $Oxyz$ rotates with the plane of the wire: $\mathbf{i}$ is horizontal along the diameter through C, $\mathbf{j}$ is vertical and upward (the rotation axis) and $\mathbf{k} = \mathbf{i} \times \mathbf{j}$ is normal to the plane, pointing out of the paper in Fig. 1a. $Ox_1y_1z_1$ is a stationary frame, assumed inertial, that coincides with $Oxyz$ at $t = 0$, with $\mathbf{j}_1 = \mathbf{j}$. The axis Oy is tangent to the wire at O, so $\overrightarrow{OC} = R\,\mathbf{i}$.

In space, P needs three coordinates. The wire removes two of them, since P has to stay in the plane Oxy and at a distance R from C, which means that the problem only has $3 - 2 = 1$ configuration degree of freedom, φ. The first constraint is reonomic because the plane moves with a known law, so the position of P seen from the stationary frame depends on φ and also explicitly on t.



(a) Plane of the wire, $\mathbf{k}$ points out of the paper. **(b)** View from above, $\mathbf{j}$ points out of the paper.

Figure 1: Geometry of the problem. The angle θ is the rotation of the plane about the vertical axis.

The angle θ turned by the plane satisfies $\dot{\theta} = \Omega$ with $\theta(0) = 0$, where $\Omega = A\omega \sin \omega t$. Integrating with

the change $u=\omega\tau$, $du=\omega\,d\tau$, and Barrow's rule:

$$\theta(t)=\int_0^t A\omega\sin(\omega\tau)\,d\tau=A\int_0^{\omega t}\sin u\,du=A\big[-\cos u\big]_0^{\omega t}=A\,(1-\cos\omega t),\tag{2.1}$$

so the plane swings between $\theta=0$ and $\theta=2A$ with period $T=2\pi/\omega$; it does not spin. For the change of basis the plane is viewed from above (Fig. 1b), with $\mathbf{j}$ pointing towards the reader. Since $\mathbf{k}_1\times\mathbf{i}_1=\mathbf{j}_1$, a positive rotation about $\mathbf{j}_1$ takes $\mathbf{k}_1$ towards $\mathbf{i}_1$ and $\mathbf{i}_1$ towards $-\mathbf{k}_1$. The vector $\mathbf{i}$ is the hypotenuse of the dashed right triangle: its leg adjacent to θ lies along $\mathbf{i}_1$ and measures $\cos\theta$, and its opposite leg lies along $-\mathbf{k}_1$ and measures $\sin\theta$, and this is where the minus sign comes from. The same triangle applied to $\mathbf{k}$ gives

$$\mathbf{i}=\cos\theta\,\mathbf{i}_1-\sin\theta\,\mathbf{k}_1,\quad \mathbf{j}=\mathbf{j}_1,\quad \mathbf{k}=\sin\theta\,\mathbf{i}_1+\cos\theta\,\mathbf{k}_1,\qquad [Q]=\begin{pmatrix}\cos\theta & 0 & \sin\theta\\ 0 & 1 & 0\\ -\sin\theta & 0 & \cos\theta\end{pmatrix},\tag{2.2}$$

where the columns of $[Q]$ are the components of $\mathbf{i},\mathbf{j},\mathbf{k}$ in the stationary basis, $\det[Q]=\cos^2\theta+\sin^2\theta=1$ and $[Q][Q]^T=I$. As a check, the chain rule gives $d\mathbf{i}/dt=-\dot\theta\,(\sin\theta\,\mathbf{i}_1+\cos\theta\,\mathbf{k}_1)=-\Omega\,\mathbf{k}$, which is $\boldsymbol{\Omega}\times\mathbf{i}$.

2.2 Position, Velocity and Acceleration of P

In the shaded right triangle at C (Fig. 1a) the hypotenuse is R, the leg adjacent to φ is $R\cos\varphi$ along $\mathbf{i}$ and the opposite one is $R\sin\varphi$ along $\mathbf{j}$. Then, with $x\equiv R(1+\cos\varphi)$ the distance from P to the rotation axis,

$$\mathbf{r}=\overrightarrow{OC}+\overrightarrow{CP}=R(1+\cos\varphi)\,\mathbf{i}+R\sin\varphi\,\mathbf{j}=x\,\mathbf{i}+R\sin\varphi\,\mathbf{j}.\tag{2.3}$$

We define the polar basis $\mathbf{e}_r=\cos\varphi\,\mathbf{i}+\sin\varphi\,\mathbf{j}$, $\mathbf{e}_\varphi=-\sin\varphi\,\mathbf{i}+\cos\varphi\,\mathbf{j}$. In $Oxyz$ the vectors $\mathbf{i},\mathbf{j}$ are constant and only φ depends on time. The chain rule, $\frac{d}{dt}\cos\varphi=-\dot\varphi\sin\varphi$ and $\frac{d}{dt}\sin\varphi=\dot\varphi\cos\varphi$, gives the relative velocity, and the product rule, $\frac{d}{dt}(\dot\varphi\sin\varphi)=\ddot\varphi\sin\varphi+\dot\varphi^2\cos\varphi$ and $\frac{d}{dt}(\dot\varphi\cos\varphi)=\ddot\varphi\cos\varphi-\dot\varphi^2\sin\varphi$, the relative acceleration:

$$\mathbf{v}_{rel}=-R\dot\varphi\sin\varphi\,\mathbf{i}+R\dot\varphi\cos\varphi\,\mathbf{j}=R\dot\varphi\,\mathbf{e}_\varphi,\tag{2.4}$$

$$\mathbf{a}_{rel}=-R(\ddot\varphi\sin\varphi+\dot\varphi^2\cos\varphi)\,\mathbf{i}+R(\ddot\varphi\cos\varphi-\dot\varphi^2\sin\varphi)\,\mathbf{j}=R\ddot\varphi\,\mathbf{e}_\varphi-R\dot\varphi^2\,\mathbf{e}_r.\tag{2.5}$$

For the motion of the frame, O is fixed, $\boldsymbol{\Omega}=\Omega\,\mathbf{j}$ and $\dot{\boldsymbol{\Omega}}=\dot\Omega\,\mathbf{j}$ with $\dot\Omega=A\omega^2\cos\omega t$, because $\mathbf{j}$ does not change. With $\mathbf{j}\times\mathbf{i}=-\mathbf{k}$, $\mathbf{j}\times\mathbf{j}=\mathbf{0}$ and $\mathbf{j}\times\mathbf{k}=\mathbf{i}$:

$$\begin{aligned}\boldsymbol{\Omega}\times\mathbf{r}&=\Omega\,\mathbf{j}\times(x\,\mathbf{i}+R\sin\varphi\,\mathbf{j})=-\Omega x\,\mathbf{k},\qquad \dot{\boldsymbol{\Omega}}\times\mathbf{r}=-\dot\Omega x\,\mathbf{k},\\ \boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\mathbf{r})&=\Omega\,\mathbf{j}\times(-\Omega x\,\mathbf{k})=-\Omega^2 x\,\mathbf{i},\\ 2\boldsymbol{\Omega}\times\mathbf{v}_{rel}&=2\Omega R\dot\varphi\,\mathbf{j}\times(-\sin\varphi\,\mathbf{i}+\cos\varphi\,\mathbf{j})=2\Omega R\dot\varphi\sin\varphi\,\mathbf{k}.\end{aligned}\tag{2.6}$$

The composition of motions [3, Eqs. 3.26–3.27] gives the velocity and acceleration in the stationary frame:

$$\mathbf{v}=\mathbf{v}_{rel}+\boldsymbol{\Omega}\times\mathbf{r}=R\dot\varphi\,\mathbf{e}_\varphi-\Omega x\,\mathbf{k},\tag{2.7}$$

$$\mathbf{a}=\big[-R(\ddot\varphi\sin\varphi+\dot\varphi^2\cos\varphi)-\Omega^2 x\big]\mathbf{i}+R(\ddot\varphi\cos\varphi-\dot\varphi^2\sin\varphi)\,\mathbf{j}+\big[2\Omega R\dot\varphi\sin\varphi-\dot\Omega x\big]\mathbf{k}.\tag{2.8}$$

2.3 Forces on the Particle

Two real forces act on P: its weight and the reaction of the wire. The wire is smooth, so the reaction has no component along the tangent $\mathbf{e}_\varphi$:

$$m\mathbf{g}=-mg\,\mathbf{j}=-mg\,(\sin\varphi\,\mathbf{e}_r+\cos\varphi\,\mathbf{e}_\varphi),\qquad \mathbf{N}=N_1\,\mathbf{e}_r+N_2\,\mathbf{k}.\tag{2.9}$$

Since $Oxyz$ is not inertial, we have to add the inertia forces [3, Eqs. 4.62–4.63]. From Eq. (2.6) and $\mathbf{i}=\cos\varphi\,\mathbf{e}_r-\sin\varphi\,\mathbf{e}_\varphi$:

$$\mathbf{F}_{cent}=-m\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\mathbf{r})=m\Omega^2 x\,\mathbf{i}=mA^2\omega^2R\sin^2(\omega t)(1+\cos\varphi)\,[\cos\varphi\,\mathbf{e}_r-\sin\varphi\,\mathbf{e}_\varphi],\tag{2.10}$$

$$\mathbf{F}_{Euler}=-m\dot{\boldsymbol{\Omega}}\times\mathbf{r}=m\dot\Omega x\,\mathbf{k}=mA\omega^2R\cos(\omega t)(1+\cos\varphi)\,\mathbf{k},\tag{2.11}$$

$$\mathbf{F}_{Cor}=-2m\boldsymbol{\Omega}\times\mathbf{v}_{rel}=-2mA\omega R\dot\varphi\sin(\omega t)\sin\varphi\,\mathbf{k}.\tag{2.12}$$

2.4 Equation of Motion and Reaction Forces

In $Oxyz$, $m\mathbf{a}_{rel} = m\mathbf{g} + \mathbf{N} + \mathbf{F}_{cent} + \mathbf{F}_{Euler} + \mathbf{F}_{Cor}$. The direction $\mathbf{e}_\varphi$ contains no reaction, and the Euler and Coriolis forces are along $\mathbf{k}$, so projecting on $\mathbf{e}_\varphi$ gives $mR\ddot{\varphi} = -mg\cos\varphi - m\Omega^2 x\sin\varphi$, that is

$$\boxed{\ddot{\varphi} = -\frac{g}{R}\cos\varphi - A^2\omega^2\sin^2(\omega t)\,(1+\cos\varphi)\sin\varphi} \tag{2.13}$$

The other two projections give the reactions. On $\mathbf{e}_r$, $-mR\dot{\varphi}^2 = N_1 - mg\sin\varphi + m\Omega^2 x\cos\varphi$, and on $\mathbf{k}$, $0 = N_2 + m\dot{\Omega}x - 2m\Omega R\dot{\varphi}\sin\varphi$:

$$N_1 = -mR\dot{\varphi}^2 + mg\sin\varphi - mA^2\omega^2 R\sin^2(\omega t)(1+\cos\varphi)\cos\varphi, \tag{2.14}$$

$$N_2 = mA\omega R\big[2\dot{\varphi}\sin(\omega t)\sin\varphi - \omega\cos(\omega t)(1+\cos\varphi)\big]. \tag{2.15}$$

All the terms are homogeneous: $[g/R] = [\Omega^2] = \mathrm{s}^{-2} = [\ddot{\varphi}]$ in Eq. (2.13), and $[mR\dot{\varphi}^2] = [mg] = [mA\omega^2 R] = \mathrm{N}$ in the reactions. Equations (2.13)–(2.15) agree with the official key of Problem 34 [2] and with the slides [4]. For $A = 0$, Eq. (2.13) is a pendulum with its stable equilibrium at the bottom, $\varphi = -\pi/2$, and the unstable one at the top, $\varphi = \pi/2$.

2.5 Mechanical Energy and Work of the Wire

With Eq. (2.7), where $\mathbf{e}_\varphi \perp \mathbf{k}$, and taking the height of O as the reference,

$$E_k = \tfrac{1}{2}m\big(R^2\dot{\varphi}^2 + \Omega^2 x^2\big), \qquad E_p = mgR\sin\varphi, \qquad E = E_k + E_p. \tag{2.16}$$

By the work–energy theorem $dE_k/dt = (m\mathbf{g}+\mathbf{N})\cdot\mathbf{v}$, and the weight is conservative, $m\mathbf{g}\cdot\mathbf{v} = -dE_p/dt$. Only the power of the reaction is left, and since $\mathbf{e}_r \perp \mathbf{e}_\varphi$ and $\mathbf{e}_r \perp \mathbf{k}$,

$$\frac{dE}{dt} = \mathbf{N}\cdot\mathbf{v} = (N_1\mathbf{e}_r + N_2\mathbf{k})\cdot(R\dot{\varphi}\,\mathbf{e}_\varphi - \Omega x\,\mathbf{k}) = -\Omega x N_2. \tag{2.17}$$

On a fixed wire the reaction is perpendicular to the velocity. Here the wire moves with the plane, so N_2 does work and E is conserved if and only if $A = 0$. From Eq. (2.15), using $\dot{x} = -R\dot{\varphi}\sin\varphi$, the torque of the wire about the axis is

$$-xN_2 = 2m\Omega x\dot{x} + m\dot{\Omega}x^2 = \frac{d}{dt}\big(m\Omega x^2\big), \tag{2.18}$$

the rate of change of the angular momentum of P about the axis, $L = m\Omega x^2$, so that $dE/dt = \Omega\,dL/dt$. In $Oxyz$ the energy includes the centrifugal potential instead, and it changes only because Ω varies:

$$J = E - \Omega L = \tfrac{1}{2}mR^2\dot{\varphi}^2 + E_p - \tfrac{1}{2}m\Omega^2 x^2, \qquad \frac{dJ}{dt} = -\dot{\Omega}L, \qquad E = J + m\Omega^2 x^2. \tag{2.19}$$

2.6 Fixed Loop: Criterion for the Motion

For $A = 0$, multiplying Eq. (2.13) by $\dot{\varphi}$ gives $\dot{\varphi}\ddot{\varphi} + (g/R)\cos\varphi\,\dot{\varphi} = 0$. By the chain rule this is a total derivative, $\frac{d}{dt}\big[\frac{1}{2}\dot{\varphi}^2 + \frac{g}{R}\sin\varphi\big] = 0$, and applying Barrow's rule between 0 and t:

$$\tfrac{1}{2}\dot{\varphi}^2 + \frac{g}{R}\sin\varphi = \tfrac{1}{2}\dot{\varphi}_0^2 + \frac{g}{R}\sin\varphi_0 = \frac{E_0}{mR^2}. \tag{2.20}$$

The potential energy has its maximum, mgR, at the top of the loop, $\varphi = \pi/2$. If $E_0 > mgR$, P reaches the top with $\dot{\varphi} \neq 0$, $\dot{\varphi}$ never changes sign and φ grows (or decreases) steadily. If $E_0 < mgR$, $\dot{\varphi}$ vanishes before the top and φ oscillates. Imposing $E_0 > mgR$ in Eq. (2.20), φ rotates if

$$|\dot{\varphi}_0| > \dot{\varphi}_c(\varphi_0) = \sqrt{\frac{2g}{R}\big(1-\sin\varphi_0\big)}, \tag{2.21}$$

with $\dot{\varphi}_0 > 0$ for a growing φ. Otherwise $\dot{\varphi} = 0$ in Eq. (2.20) gives the turning points, $\varphi_{max} = \arcsin\big(\sin\varphi_0 + R\dot{\varphi}_0^2/(2g)\big)$ and $\varphi_{min} = -\pi - \varphi_{max}$. The limit $E_0 = mgR$ is the separatrix $\dot{\varphi} = \pm\sqrt{(2g/R)(1-\sin\varphi)}$, along which the top is only approached in an infinite time: near it $\epsilon = \pi/2 - \varphi$ follows $\dot{\epsilon} \approx -\sqrt{g/R}\,\epsilon$.

2.7 Linearized Equation

With $\varphi = \delta - \pi/2$, $\cos\varphi = \sin\delta$, $\sin\varphi = -\cos\delta$ and $\ddot{\varphi} = \ddot{\delta}$, so Eq. (2.13) becomes the exact $\ddot{\delta} = -(g/R)\sin\delta + \Omega^2(1+\sin\delta)\cos\delta$. For $\delta \ll 1$, $\sin\delta \approx \delta$ and $\cos\delta \approx 1$, which gives $\ddot{\delta} = -(g/R)\delta +$

$\Omega^2(1+\delta)$. The hypothesis $A^2 \ll g/(R\omega^2)$ means $\Omega^2 \le A^2\omega^2 \ll g/R$, so $\Omega^2\delta$ is dropped against $(g/R)\delta$ [4]. Using $\sin^2\omega t = \frac{1}{2}(1-\cos 2\omega t)$:

$$\ddot\delta + \frac{g}{R}\,\delta - \Omega^2 = 0 \quad\Longleftrightarrow\quad \ddot\delta + \omega_n^2\delta = F_0\,(1-\cos 2\omega t), \qquad \omega_n = \sqrt{g/R}, \quad F_0 = \tfrac{1}{2}A^2\omega^2. \tag{2.22}$$

This is an undamped oscillator forced at the frequency 2ω by the centrifugal force, which peaks twice per period of the plane. The guidelines give no initial conditions, so we take P at rest at the bottom, $\delta(0) = \dot\delta(0) = 0$, and then all the motion comes from the loop. The constant F_0/ω_n^2 and $C\cos 2\omega t$ with $C = -F_0/(\omega_n^2 - 4\omega^2)$ are a particular solution; adding $a\cos\omega_n t + b\sin\omega_n t$ and imposing the initial conditions:

$$\delta = \frac{F_0}{\omega_n^2} - \frac{F_0\cos 2\omega t}{\omega_n^2 - 4\omega^2} + \Big(\frac{F_0}{\omega_n^2 - 4\omega^2} - \frac{F_0}{\omega_n^2}\Big)\cos\omega_n t, \qquad 2\omega \ne \omega_n. \tag{2.23}$$

Thus P oscillates around $\delta_s = F_0/\omega_n^2 = A^2R\omega^2/(2g)$, displaced outwards from the bottom. If $2\omega = \omega_n$ the denominator vanishes. This is resonance [3, Sec. 5.4.1], and the particular solution changes to $-\frac{F_0}{2\omega_n}t\sin\omega_n t$, which satisfies $\ddot\delta + \omega_n^2\delta = -F_0\cos\omega_n t$:

$$\delta = \frac{F_0}{\omega_n^2}(1-\cos\omega_n t) - \frac{F_0}{2\omega_n}\,t\sin\omega_n t, \qquad 2\omega = \omega_n. \tag{2.24}$$

The amplitude grows linearly with time, so $\delta \ll 1$ stops holding at $\omega = \frac{1}{2}\sqrt{g/R}$.

3 Numerical Implementation in Matlab

Equation (2.13) has to be written for the state vector $X = [\varphi, \dot\varphi]^T = [X_1, X_2]^T$ because `ode45` [5] only integrates first-order systems:

$$\frac{d}{dt}\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = f(t,X) = \begin{pmatrix} X_2 \\ -\dfrac{g}{R}\cos X_1 - A^2\omega^2\sin^2(\omega t)\,(1+\cos X_1)\sin X_1 \end{pmatrix}. \tag{3.1}$$

This is the function `diffeq(t,X,g,omega,R,A)` of `main.m`, and Eq. (2.22) is `diffeqlin`, with $X = [\delta, \dot\delta]^T$. The options are `RelTol` $= 10^{-8}$, `AbsTol` $= 10^{-10}$ (the default 10^{-6} is not small against $\delta \sim 10^{-3}$) and `MaxStep` $= 0.01$ s, at least 237 steps per swing, far above the Nyquist limit [1]. We compute the forces and energies afterwards from φ and $\dot\varphi$ with Eqs. (2.10)–(2.15) and (2.16), and the work of the wire W_N with the trapezoidal rule applied to Eq. (2.17). The position in the stationary frame follows from Eqs. (2.1)–(2.3): $x_1 = x\cos\theta$, $y_1 = R\sin\varphi$, $z_1 = -x\sin\theta$. A fixed-loop case counts as rotating if $\dot\varphi$ never changes sign and as oscillating otherwise, and the periods come from the times of the extrema of φ. All the numbers in this report are printed by `main.m`.

In all cases $g = 9.81\,\mathrm{m/s^2}$ and $m = 1\,\mathrm{kg}$. The oscillating loop uses $A = \pi$, $R = 1\,\mathrm{m}$, $\omega = 0.1\,\mathrm{rad/s}$, $\varphi(0) = \dot\varphi(0) = 0$ and $t \in [0, 3T]$ with $T = 62.83\,\mathrm{s}$. The fixed loop uses $A = 0$ and $R = 1\,\mathrm{m}$, and the linearized study $A = 0.1$, $R = 0.01\,\mathrm{m}$ and $\omega_n = 31.32\,\mathrm{rad/s}$. The guidelines do not fix the duration of these two studies, so we take $t \in [0, 10]$ s, longer than any period of the fixed loop and about 50 natural periods of the linearized study.

4 Results and Discussion

4.1 Angle and Reaction Forces

Figure 2 shows φ, N_1 and N_2 over $[0, 3T]$ and, since the 79.5 swings of that interval cannot be seen one by one, a window of 10 s around $t = T/4$, when the plane turns fastest. At $t = 0$ the plane is at rest and Eq. (2.13) gives $\ddot\varphi(0) = -g/R = -9.81\,\mathrm{rad/s^2}$: P falls from the end of the horizontal diameter and swings around the bottom between $\varphi \approx 0$ and $\varphi \approx -\pi$, the point O. The amplitude does not change over the three periods although $A = \pi$ is a large rotation, because what enters Eq. (2.13) is $A\omega$: Ω^2 is at most $(A\omega)^2$, 1.01% of g/R. So P behaves as a pendulum of 90° amplitude. Its swing period, 2.370 s, is longer than the small-oscillation value $2\pi\sqrt{R/g} = 2.006\,\mathrm{s}$ and close to the exact 2.368 s of that pendulum. The highest angle, 0.0118 rad, puts P 0.0118 m above its start, using energy given by the wire (Section 4.3).

N_1 stays between $-29.43\,\mathrm{N}$ and $0.00\,\mathrm{N}$, so the wire pulls P towards C. It vanishes at the turning

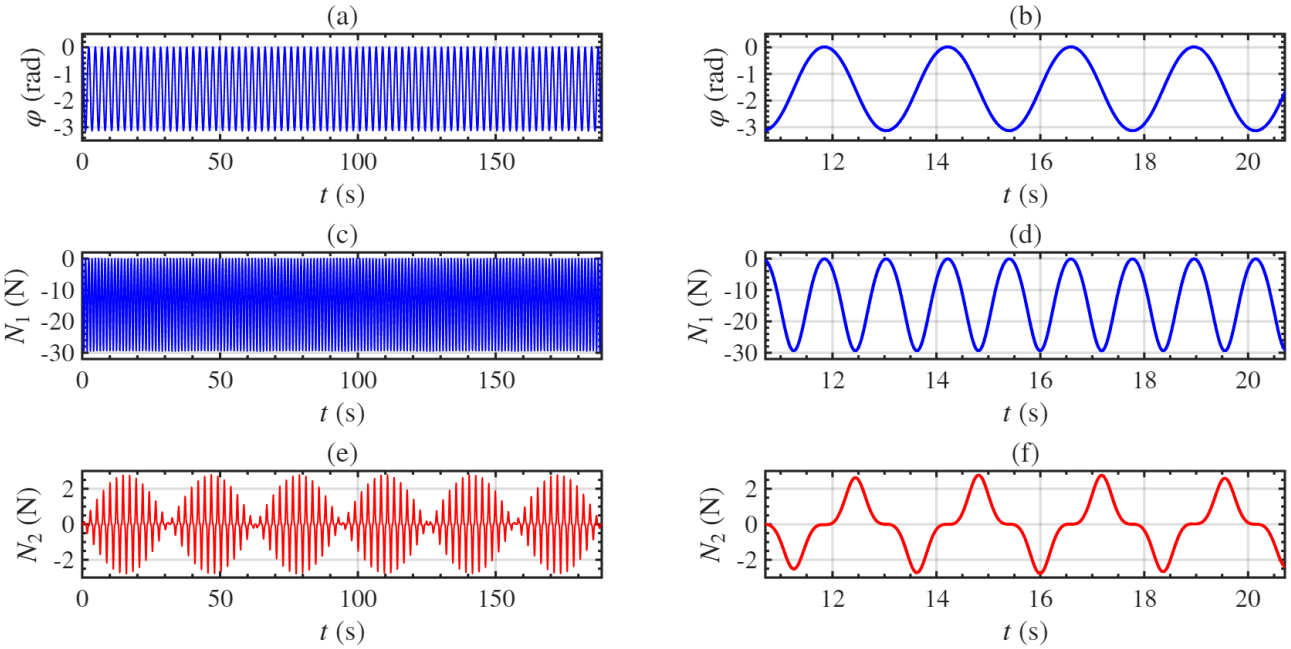


Figure 2: Oscillating loop. Left: $t \in [0, 3T]$. Right: window around $t = T/4$. (a, b) Angle; (c, d) reaction along $\mathbf{e}_r$; (e, f) reaction along $\mathbf{k}$.

points, where $\dot{\varphi} = 0$ and $\sin\varphi \approx 0$, and its minimum is at the bottom: since the rotation is weak, Eq. (2.20) holds approximately and gives $\dot{\varphi}^2 \approx 2g/R$, and Eq. (2.14) gives $N_1 \approx -2mg - mg = -3mg$, the weight plus the force that makes P turn. As this happens at every crossing of the bottom, N_1 oscillates twice as fast as φ (Fig. 2d). N_2 is 10.6 times smaller, between $-2.777\,\mathrm{N}$ and $2.778\,\mathrm{N}$. Its envelope has period $T/2$ and falls to the Euler force alone when the plane stops and turns back. Inside a swing it peaks at the bottom, where $\dot{\varphi}$ is largest, and its sign alternates between consecutive crossings because P crosses in opposite directions (Fig. 2f). We explain this in the next section.

4.2 Inertia Forces

The three forces have different sizes, so each has its own axis in Fig. 3, as the guidelines require. F_{cent} always points away from the axis. Its peak, $0.1974\,\mathrm{N}$, is $2mA^2\omega^2R$, reached with P at $x = 2R$ when $|\Omega|$ is maximum, and it is only 2.01% of the weight. F_{Euler} follows $\cos\omega t$ and is largest ($0.0628\,\mathrm{N}$) when the plane stops and F_{cent} and F_{Cor} vanish, since it depends on the change of Ω and not on its value. F_{Cor} is the largest, up to $2.778\,\mathrm{N}$, 44.2 times F_{Euler}, because P moves fast along the wire ($|\dot{\varphi}| \le 4.430\,\mathrm{rad/s}$) compared with the slow plane ($\omega = 0.1\,\mathrm{rad/s}$).

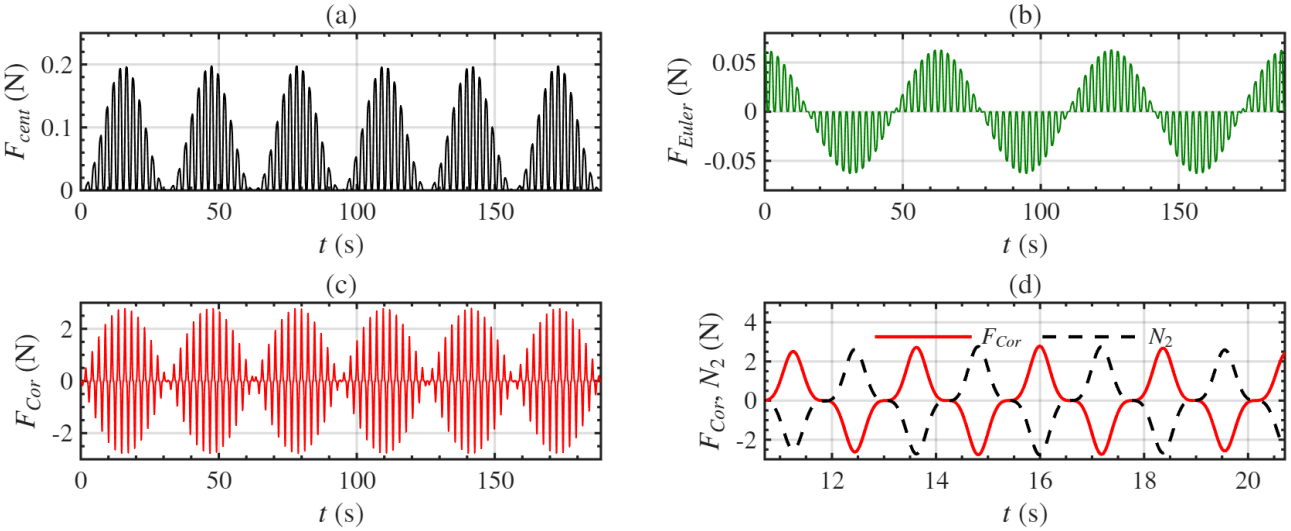


Figure 3: Inertia forces of the oscillating loop. (a) Centrifugal (along $\mathbf{i}$), (b) Euler and (c) Coriolis (along $\mathbf{k}$) for $t \in [0, 3T]$. (d) Coriolis force and reaction N_2 around $t = T/4$.

Equation (2.15) is $N_2 = -F_{Cor} - F_{Euler}$, so the wire has to balance both, and since the Euler force is

small N_2 is almost the mirror image of F_{Cor} (Fig. 3d). This means that the largest inertia force does not change φ. It is normal to the plane of motion, so it only loads the wire, and through N_2 it sets the energy exchange of Section 4.3. The only inertia force in Eq. (2.13) is the part of the centrifugal one along $\mathbf{e}_\varphi$, which is why the swing of Fig. 2a is almost unaffected by the rotation.

4.3 Energy

In Fig. 4(a, b) $E_k \in [0.000, 9.830]$ J and $E_p \in [-9.810, 0.115]$ J exchange about 9.8 J in each half swing, as in any pendulum. Their sum is not constant: $E - E_0$ moves between -0.0824 J and 0.3129 J (Fig. 4c), as Eq. (2.17) predicts. The work W_N computed from the power $-\Omega x N_2$ reproduces $E - E_0$ within 2.4×10^{-6} J, so we can trust N_2, E and the integration.

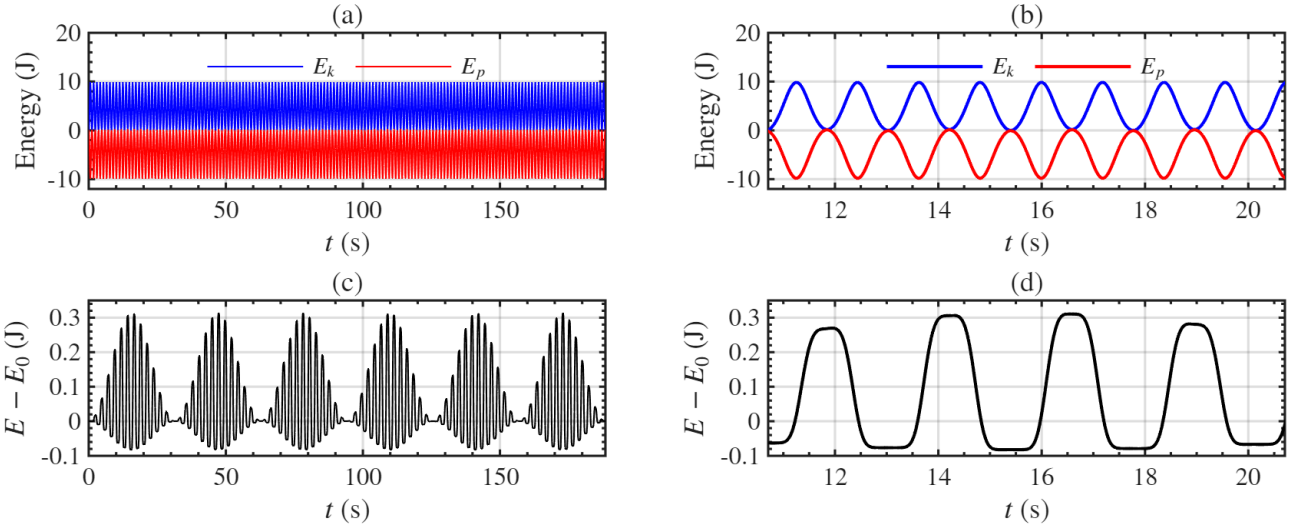


Figure 4: Energy of the oscillating loop. (a, b) Kinetic and gravitational potential energy; (c, d) change of the mechanical energy. Left: $t \in [0, 3T]$. Right: window around $t = T/4$.

In Fig. 4d we can see the reason: E is high while P is far from the axis and low while it is near. While P moves outwards at an almost constant Ω, the wire must increase $L = m\Omega x^2$ and the motor gives the energy $\Omega\,\Delta L$ (Eq. (2.18)); it takes it back when P returns. From $x = 0$ to $x = 2R$ at the maximum Ω this is $4mA^2\omega^2R^2 = 0.3948$ J, and the range of E in the simulation is 0.3953 J. E also falls below E_0, and this is explained by Eq. (2.19). As Ω varies slowly compared with the swing, J follows $-\frac{1}{2}m\Omega^2\langle x^2\rangle$, with $\langle x^2\rangle = 1.670\,\mathrm{m}^2$ the average over the swing. At the maximum Ω this predicts $E - E_0 = -0.0824$ J at $x = 0$ and 0.3124 J at $x = 2R$, which is what we get in the simulation. When the plane stops, $\Omega = 0$ and J returns to almost zero, so there is no net gain after each of its periods (26.5 swings): $E - E_0$ is 0.0006 J at T, 0.0000 J at $2T$ and 0.0006 J at $3T$. This is why the amplitude of Fig. 2a does not drift.

4.4 Trajectory in the Stationary Frame

From the ground (Fig. 5), P moves on the surface swept by the wire, a torus without a hole because the circle touches the axis: the distance to the axis goes from practically zero (1.0×10^{-12} m) to 2.0000 m. Each swing draws one petal from the axis to $x = 2R$ and back, and the plane spreads them around the axis because it turns up to $\theta = 2A = 2\pi$ at $t = T/2$ and back to 0 at $t = T$. The petals are not evenly spaced because of the law of Ω. Near $\theta = 0$ and $\theta = 2\pi$ the plane is almost stopped, so the petals bunch around $+x_1$ (Fig. 5b); near $\theta = \pi$ it turns fastest, up to 0.744 rad per swing, and they open up. The three periods give almost the same flower, slightly displaced because T is not a multiple of the swing period. The video `trajectory_video.avi` (generated by `main.m`) shows the first period with the wire moving.

4.5 Fixed Loop: Phase Portrait and Validation of the Criterion

Figure 6 shows the nine initial conditions together with the separatrix of Eq. (2.21), and Table 1 compares the analytical prediction with the simulation. φ is plotted modulo 2π, so both ends of the axis are the top of the loop. The prediction and the simulation agree in all nine cases, and φ_{max} agrees

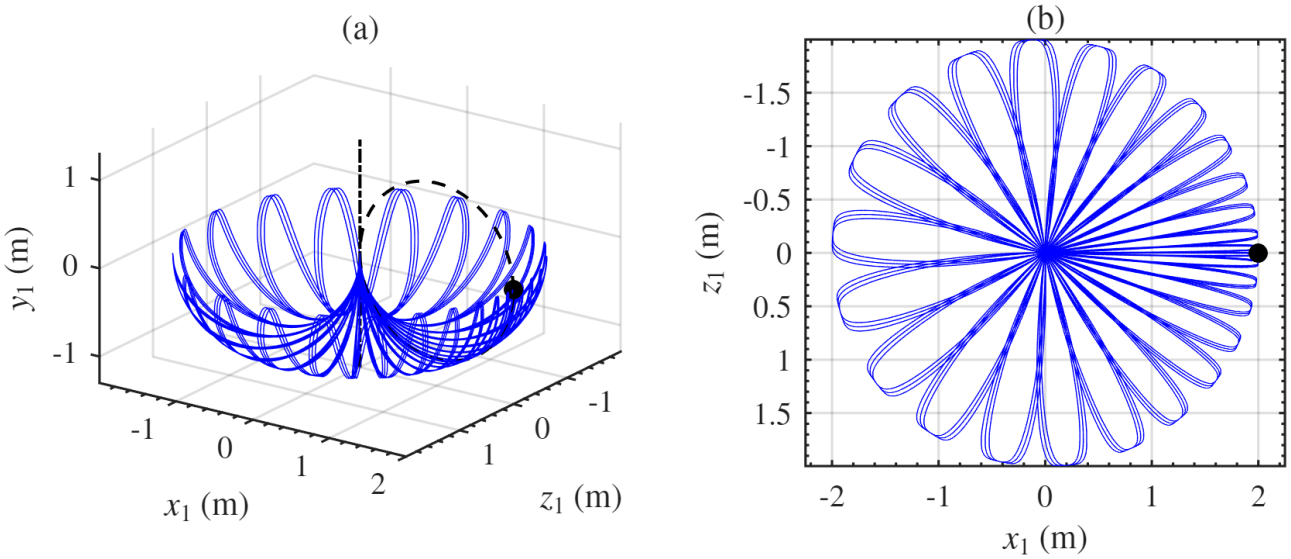


Figure 5: Trajectory of P in $Ox_1y_1z_1$ for $t \in [0, 3T]$, with the start marked by a dot. (a) Three-dimensional view with the wire at $t = 0$ (dashed) and the rotation axis (dash-dot). (b) View from above, with z_1 pointing down as in Fig. 1b.

with the formula of Section 2.6 to all the digits shown. As Eq. (2.20) requires, E is conserved, here within $4.0 \times 10^{-10}\ mgR$, which also shows that the numerical error is negligible.

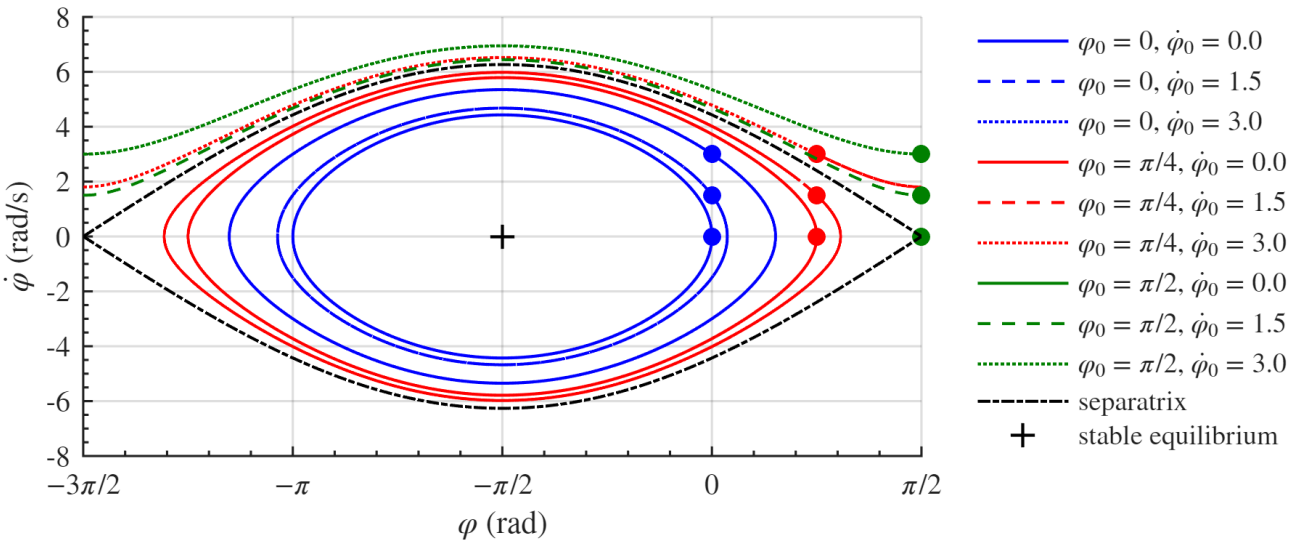


Figure 6: Phase portrait of the fixed loop ($A = 0$), with φ_0 in rad and $\dot{\varphi}_0$ in rad/s. Dots mark the initial conditions and the dash-dot line is the separatrix. Closed curves are oscillations and open curves are rotations.

Table 1: Criterion of Eq. (2.21) against the simulation of the fixed loop. φ_{max} and the period come from the simulation.

φ_0 (rad)	$\dot{\varphi}_0$ (rad/s)	$E_0/(mgR)$	$\dot{\varphi}_c$ (rad/s)	Prediction	Simulation	φ_{max} (rad)	Period (s)
0	0	0.000	4.429	oscillates	oscillates	0.0000	2.368
0	1.5	0.115	4.429	oscillates	oscillates	0.1149	2.434
0	3	0.459	4.429	oscillates	oscillates	0.4765	2.709
$\pi/4$	0	0.707	2.397	oscillates	oscillates	0.7854	3.065
$\pi/4$	1.5	0.822	2.397	oscillates	oscillates	0.9645	3.362
$\pi/4$	3	1.166	2.397	rotates	rotates	–	–
$\pi/2$	0	1.000	0.000	rest	rest	1.5708	–
$\pi/2$	1.5	1.115	0.000	rotates	rotates	–	–
$\pi/2$	3	1.459	0.000	rotates	rotates	–	–

Only ($\pi/4$, 3), ($\pi/2$, 1.5) and ($\pi/2$, 3) give a φ that increases steadily: they start above the separatrix, with $E_0 > mgR$, so their curves never touch $\dot{\varphi} = 0$, and P runs slowest at the top and fastest at the bottom. The speed alone does not decide it: (0, 3) starts with the same $\dot{\varphi}_0$ as ($\pi/4$, 3) but with less

potential energy, $E_0/(mgR) = 0.459$, and it oscillates. The period of the closed curves grows from 2.368 s to 3.362 s as they approach the separatrix, because P spends more and more time near the top, where it almost stops. For $(\pi/2,\, 0)$ the analysis gives rest on the equilibrium and the simulation agrees over 10 s ($|\dot{\varphi}| \le 6.0\times10^{-15}$ rad/s), but this is a numerical effect. In the computer $\cos(\pi/2) = 6.1\times10^{-17}$ is a tiny push, and at first each step of φ is below the rounding of φ. Integrating further, φ first changes at $t = 18.49$ s and P has fallen to $\varphi = 0$ at $t = 29.4$ s: the equilibrium is unstable, and a real particle would leave it with any disturbance.

4.6 Linearized Against Full Equation

Figure 7a shows the percentage error $100\,|\delta_{lin} - \delta_{full}|/\max|\delta_{full}|$. The division is by the amplitude and not by $\delta(t)$ because δ crosses zero, where the relative error would diverge. The error covers several orders of magnitude, so the axis is logarithmic [1]. It also oscillates fast, so its value at one instant does not say much and we compare its envelope. The numerical solution of Eq. (2.22) matches Eqs. (2.23)–(2.24) within 1.7×10^{-8} rad, so the differences come from the model and not from the integration. To see which approximation causes it, Table 2 also gives the error when each neglected term is restored alone in Eq. (2.22).

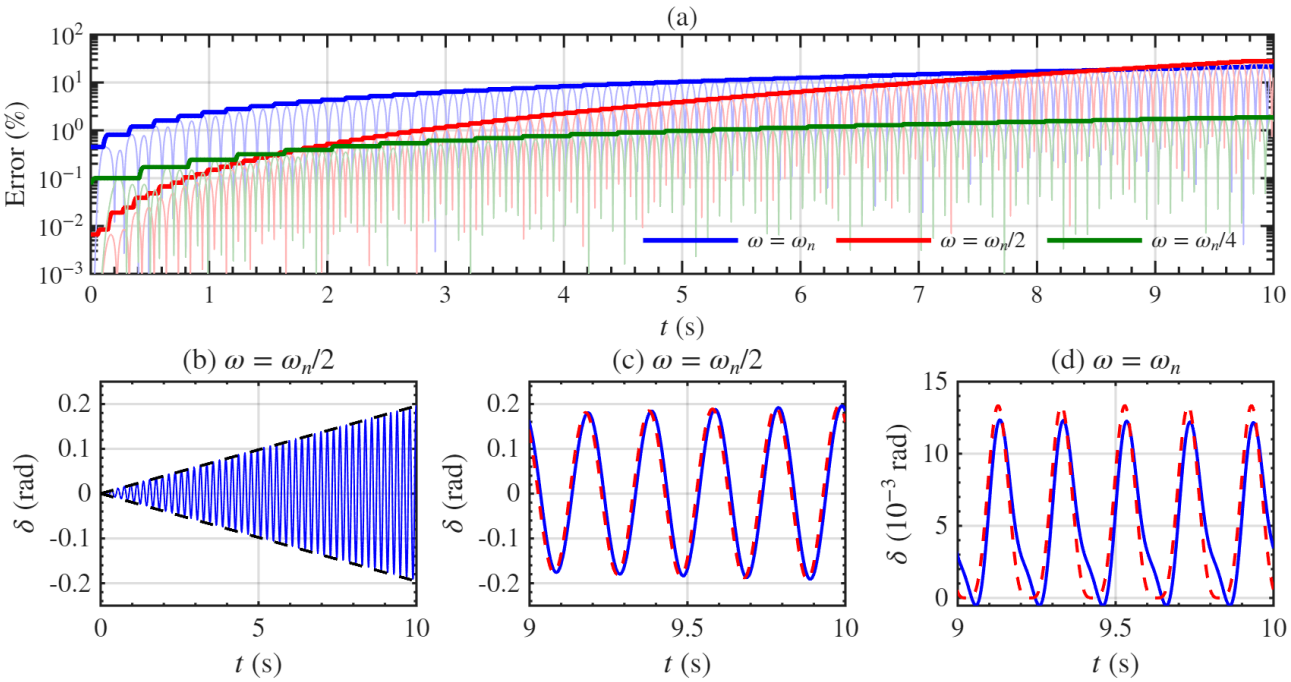


Figure 7: (a) Percentage error of the linearized solution (thin: error; thick: its envelope). (b) Full solution at resonance and envelope $\pm F_0 t/(2\omega_n)$ (dashed). (c, d) Full solution (solid) and linearized solution (dashed) in the last second.

Table 2: Linearized against full equation for $\delta(0) = \dot{\delta}(0) = 0$, with $\omega_n = 31.32$ rad/s. All errors in %.

ω	δ_s (rad)	$\max\lvert\delta_{full}\rvert$ (rad)	Largest error		Largest error in $[0, 10]$ s, keeping	
			$[0, 5]$ s	$[0, 10]$ s	$\Omega^2\delta$	$\sin\delta$
ω_n	0.00500	0.01339	10.09	21.36	0.072	21.212
$\omega_n/2$	0.00125	0.19559	3.47	27.75	18.182	9.616
$\omega_n/4$	0.00031	0.00083	0.90	1.86	0.001	1.860

At $\omega = \omega_n/2$ the forcing frequency 2ω equals ω_n: every push of the centrifugal force is in phase with the natural swing, which is the resonance of Eq. (2.24). The amplitude grows along the line $F_0 t/(2\omega_n)$, which gives 0.1958 rad at 10 s against the 0.195 59 rad of the full equation (Fig. 7b). So δ reaches 11.2°, 14.6 times the value at ω_n, although the forcing is 4 times weaker. In Fig. 7a its error starts as the smallest, partly because it is divided by this large final amplitude, and it ends as the largest because as δ grows the pendulum softens and falls out of phase with the forcing (Fig. 7c). Both approximations contribute: restoring $\sin\delta$ reduces the error from 27.75% to 9.616%, and restoring $\Omega^2\delta$ only to 18.182%. The softening also limits the resonance: integrating the full equation up to 100 s, δ

peaks at 0.324 rad at $t = 20.5$ s, while the linearized envelope reaches 1.96 rad.

At $\omega = \omega_n$, δ stays small and $\sin\delta \approx \delta$ is excellent, but the error still reaches 21.36%. All of it comes from $\Omega^2\delta$: restoring this term leaves 0.072%. This term looks negligible, since the mean of Ω^2 is $A^2\omega^2/2 = 4.905\,\mathrm{s}^{-2}$, 0.500% of g/R. However, it is a stiffness that varies in time at $2\omega = 2\omega_n$, twice the natural frequency (a parametric excitation, which goes beyond the course), and its effect on the natural oscillation builds up swing after swing. A fit of the last second shows the natural part of δ_{full} 0.41 rad behind that of δ_{lin} and with 0.875 of its amplitude (Fig. 7d), which alone gives an error of 20.0%. That is why the error doubles from 5 s to 10 s. At $\omega = \omega_n/4$ the forcing is 16 times weaker than at ω_n and the error stays below 2%, so here the linearization can be used. The three cases satisfy $A^2 = 0.01 \ll g/(R\omega^2) = 1, 4, 16$, and the error still differs by more than one order of magnitude. The hypothesis only tells us that the neglected term is small in the equation, and its effect can still grow in time.

4.7 Limits of the Model

The wire has no friction and nothing removes energy. Even so, only the linearized equation grows without limit: the full one is limited by its softening, and real damping would reduce the amplitude further [3, Sec. 5.4.1]. The rotation of the plane is imposed, which assumes a motor able to give the torque $|xN_2|$ (up to 3.422 N m here) whatever P does. The wire is bilateral: N_1 and N_2 can take either sign, so unlike the string of Lab 1 no event ends the motion. Finally, Table 2 holds for our choice of initial conditions: with $\delta(0) = 0.1$ rad the largest errors rise to 55%, 43% and 24%, because the free oscillation is larger, so the conclusion at $\omega_n/4$ is only valid for P starting at rest at the bottom.

5 Conclusions

- The particle has one degree of freedom, φ, and Eq. (2.13) contains only gravity and the centrifugal force. The Coriolis force is the largest inertia force, but it only loads the wire, through N_2. For $\omega = 0.1$ rad/s the rotation is slow, so P swings as a pendulum of 90° amplitude and the wire carries up to $3mg$ in its plane.
- As we have derived in Eq. (2.17), $dE/dt = -\Omega x N_2$. This means that the mechanical energy is not conserved, because the moving wire does work. The motor gives and takes back the energy needed to change the angular momentum about the axis, so E oscillates with each swing but has almost no net gain after each period of the plane.
- On a fixed loop, the energy gives the critical rate $\dot{\varphi}_c = \sqrt{(2g/R)(1-\sin\varphi_0)}$, and all the simulated cases confirm it. Only three of the nine initial conditions rotate.
- We have seen that the linearized equation is a forced oscillator with a resonance at $\omega = \frac{1}{2}\sqrt{g/R}$, where $\delta \ll 1$ fails. It also loses accuracy away from resonance, since a term that is small in the equation builds up in time. Only the weakest forcing, starting at rest at the bottom, keeps the error below 2%.

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